

Spatial Transcriptomics

Seeing is believing, ...



Anjaney Pandey

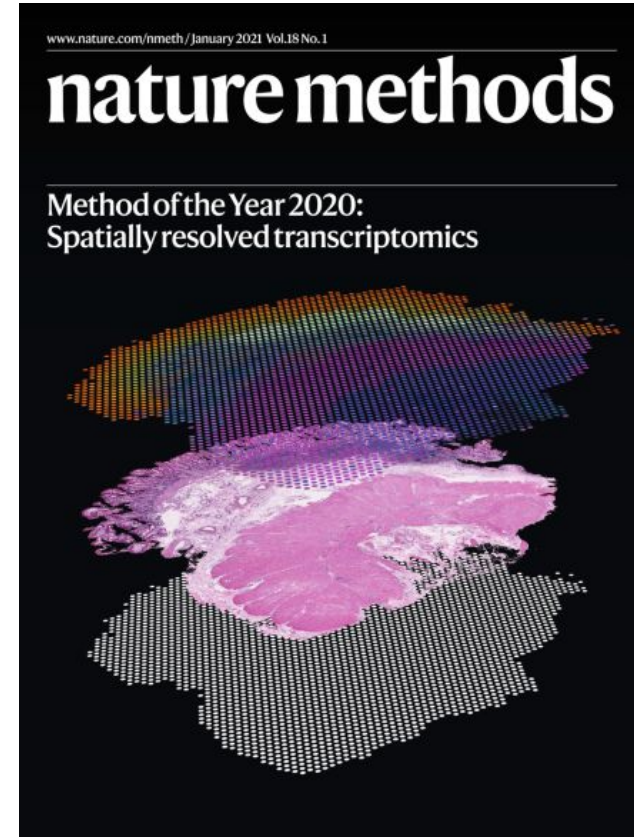
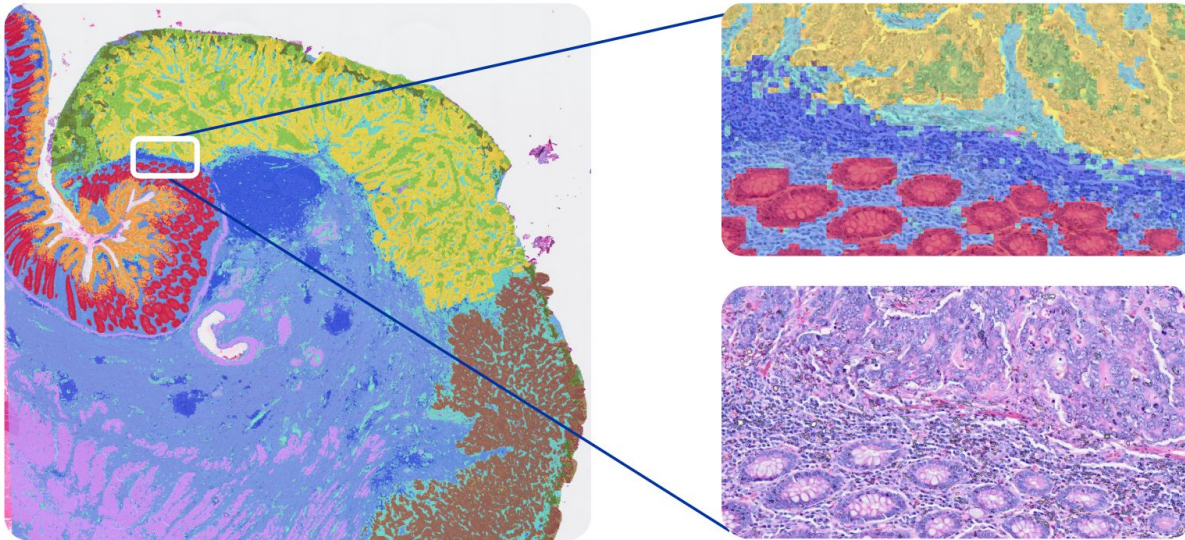
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What is spatial transcriptomics (ST)?

To capture the spatial context of gene expression within intact tissue

Combines sequencing and imaging to provide maps of gene activity



The need for single cell transcriptomics with spatial analysis

Bulk



Single cell



Single cell spatial analysis



VS

VS

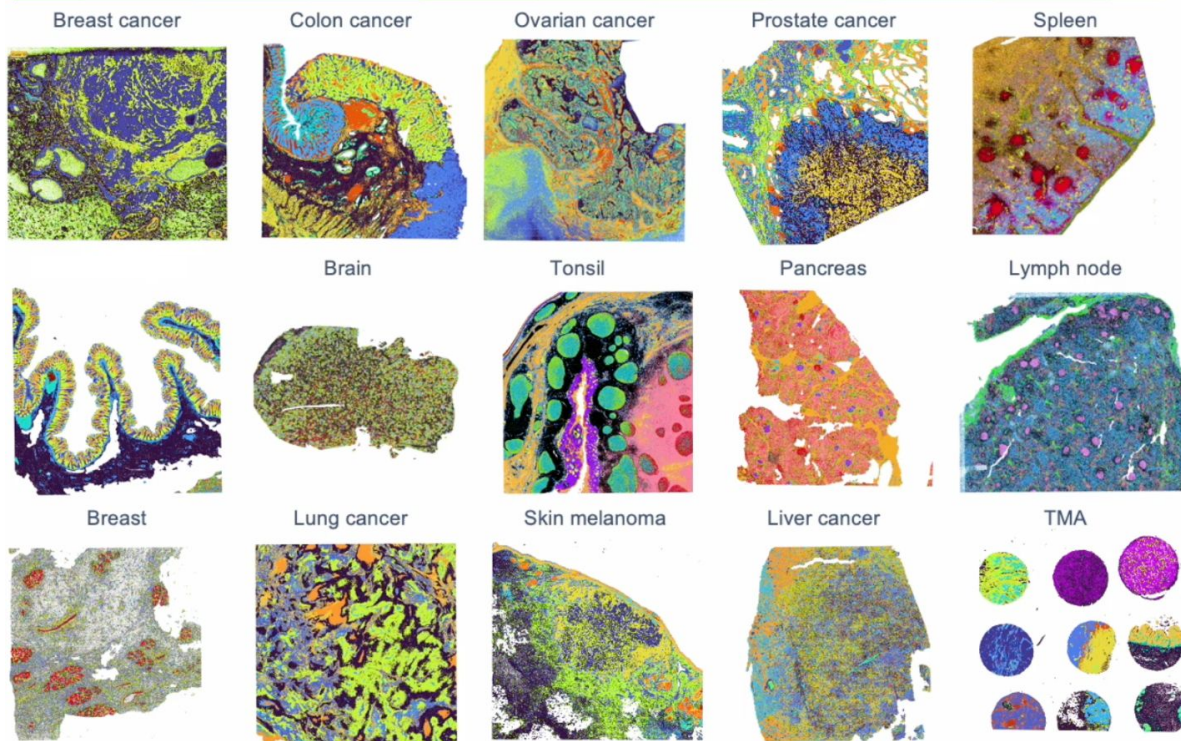
Bulk transcriptomic analyses lose spatial and single cell information

Single cell transcriptomic analyses retain single cell information but lose spatial organization

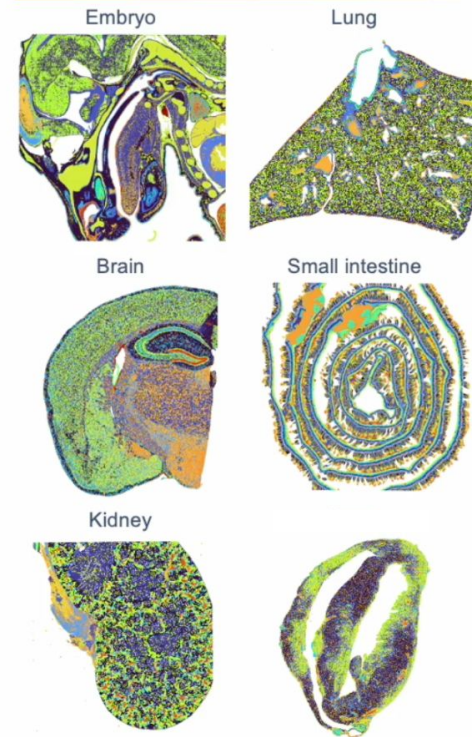
Single cell, multiplexing spatial analyses provide transcriptomic information with spatial organization while retaining single cell resolution

Tissues with spatial data

Human

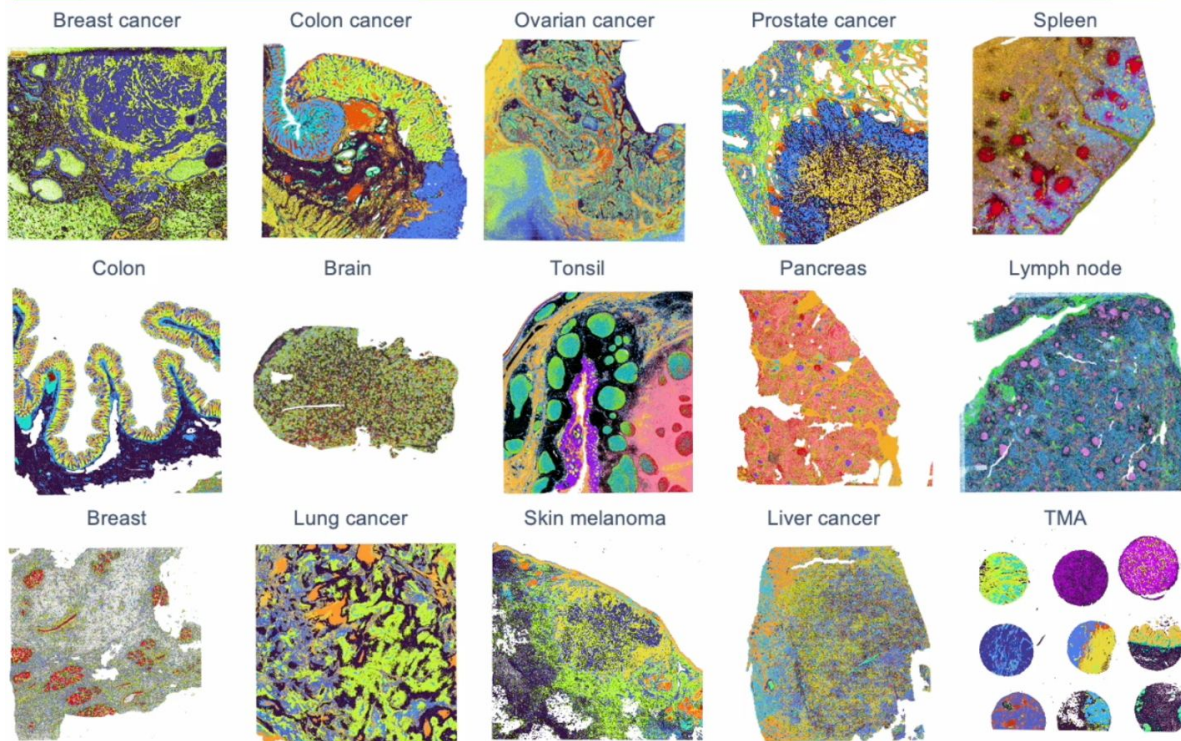


Mouse

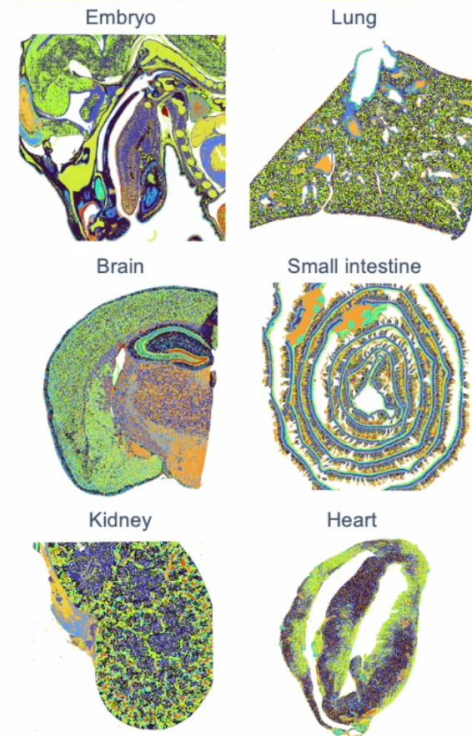


Tissues with spatial data

Human



Mouse



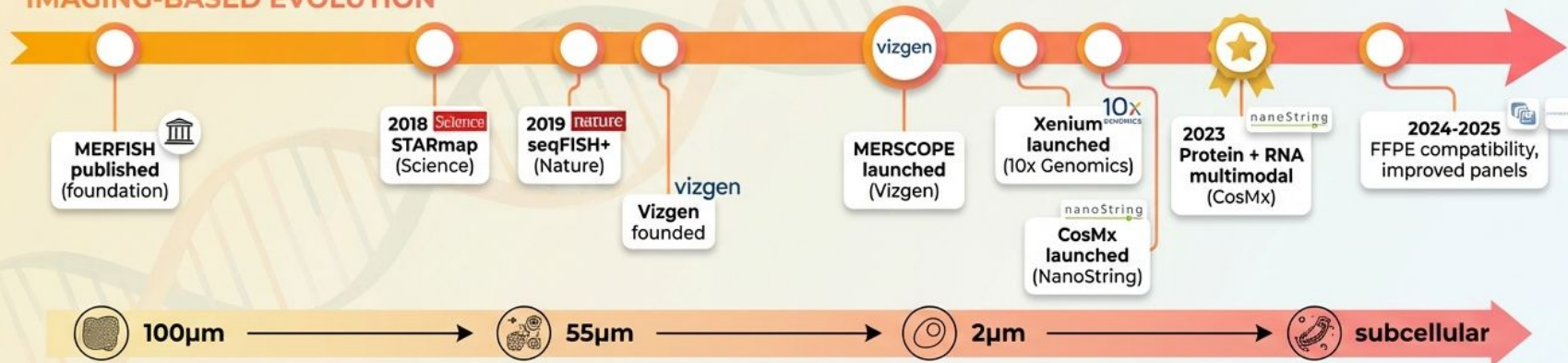
Evolution of ST

EVOLUTION OF SPATIAL TRANSCRIPTOMICS PLATFORMS (2016-2025)

SEQUENCING-BASED EVOLUTION

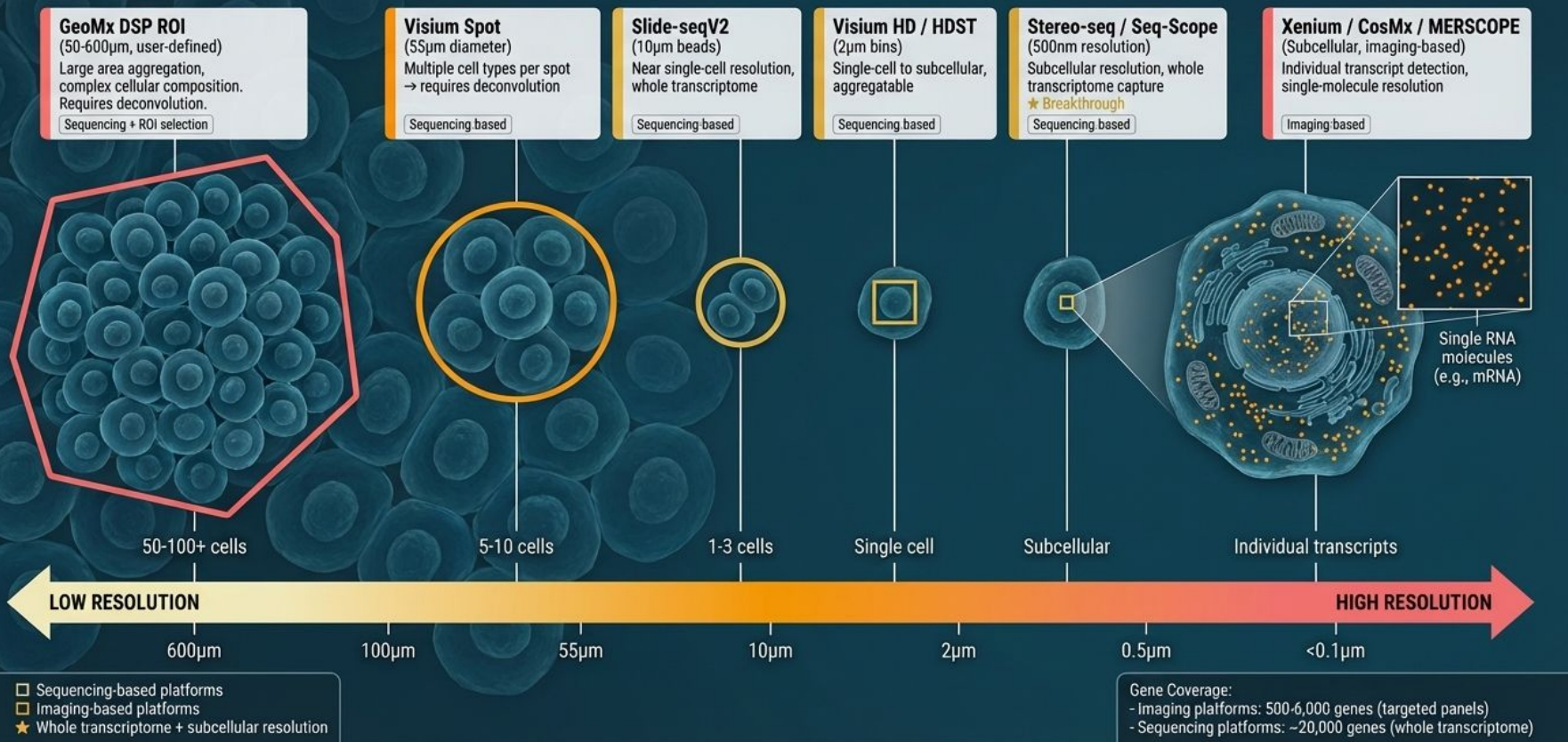


IMAGING-BASED EVOLUTION



SPATIAL RESOLUTION ACROSS TRANSCRIPTOMICS PLATFORMS

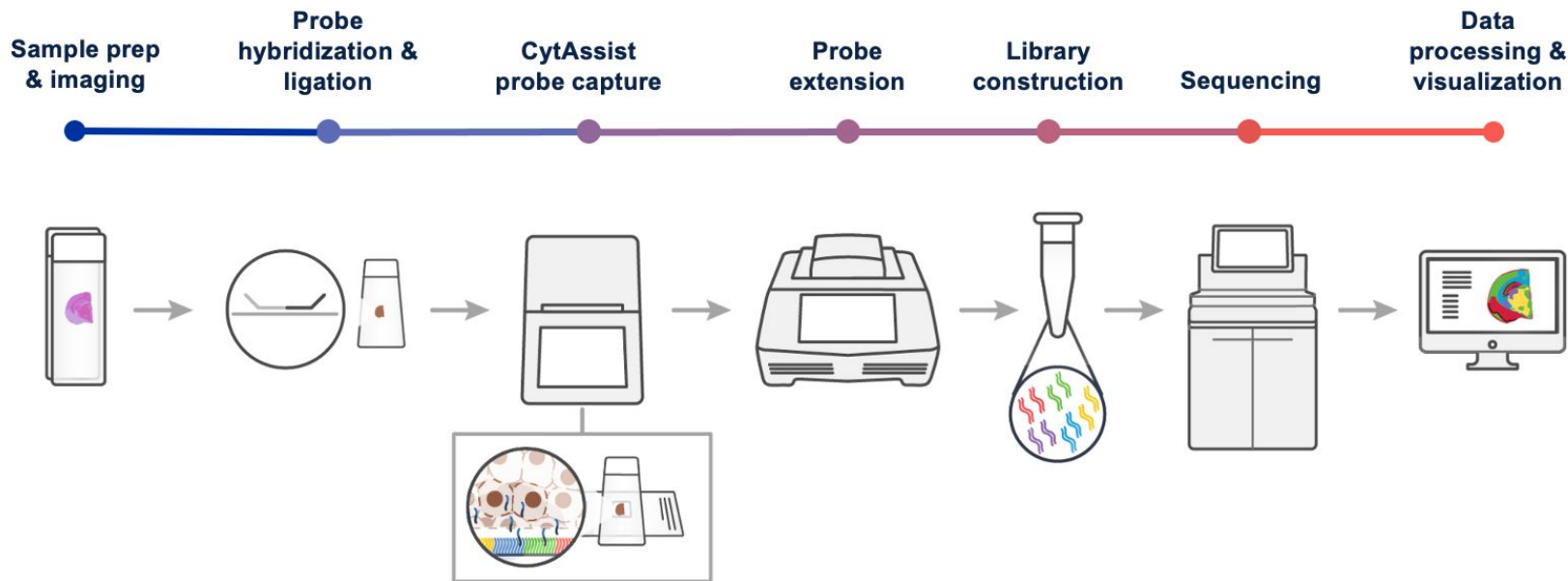
From Multi-cell Regions to Individual Transcripts

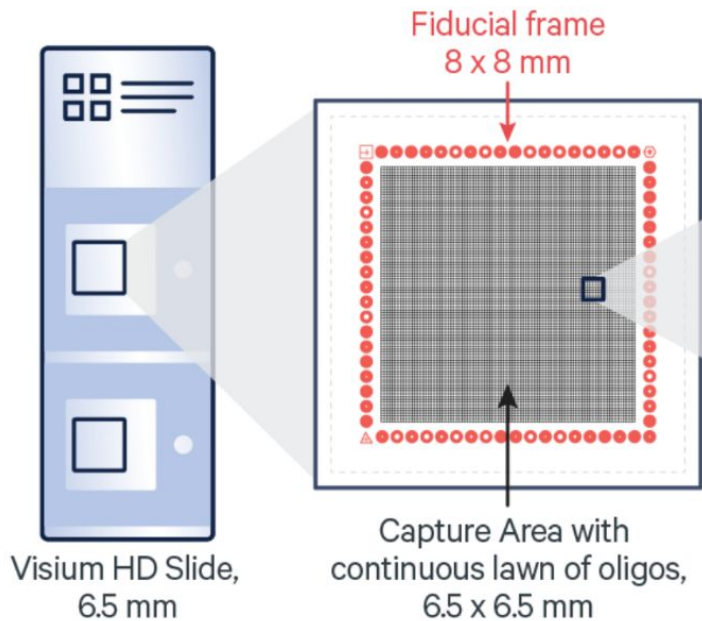


Visium HD

Visium HD Workflow Powered by CytAssist

Transforming standard histological sections into high-resolution, whole transcriptome gene expression data

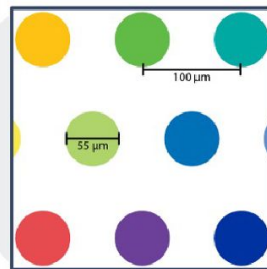




Visium

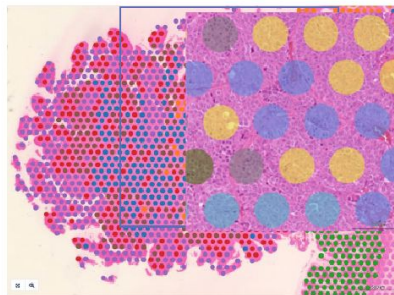


Capture Area with 55 µm barcoded spots,
their centers 100 µm apart



55 µm spots
Hexagonally arranged

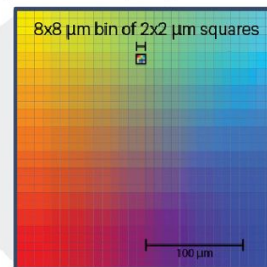
5,000
features



Visium HD

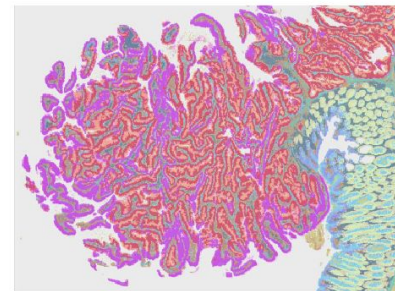


Capture Area with continuous lawn of
2x2 µm barcoded squares, binned to 8x8 µm



2 µm squares
Continuous grid-pattern

11,000,000
features

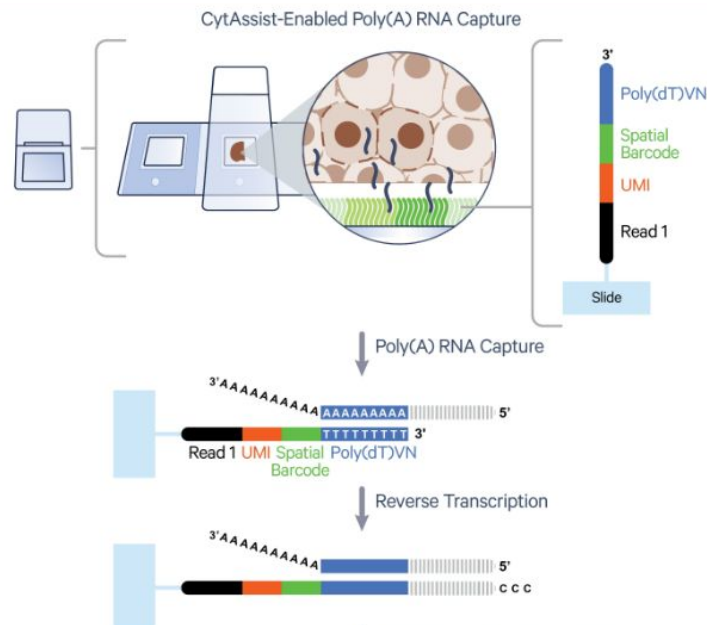


CytAssist powered experimental workflow

Facilitates transfer of transcript probes

- Sectioning, tissue preparation, staining (H&E or IF), and imaging take place on a **standard glass slide**
- After probe hybridization, two glass slides and a **Visium slide** are placed in the CytAssist instrument for alignment
- **Brightfield image** is captured to provide spatial orientation for data analysis

Followed by next-generation sequencing which generates bcl files which are converted to fastq.gz



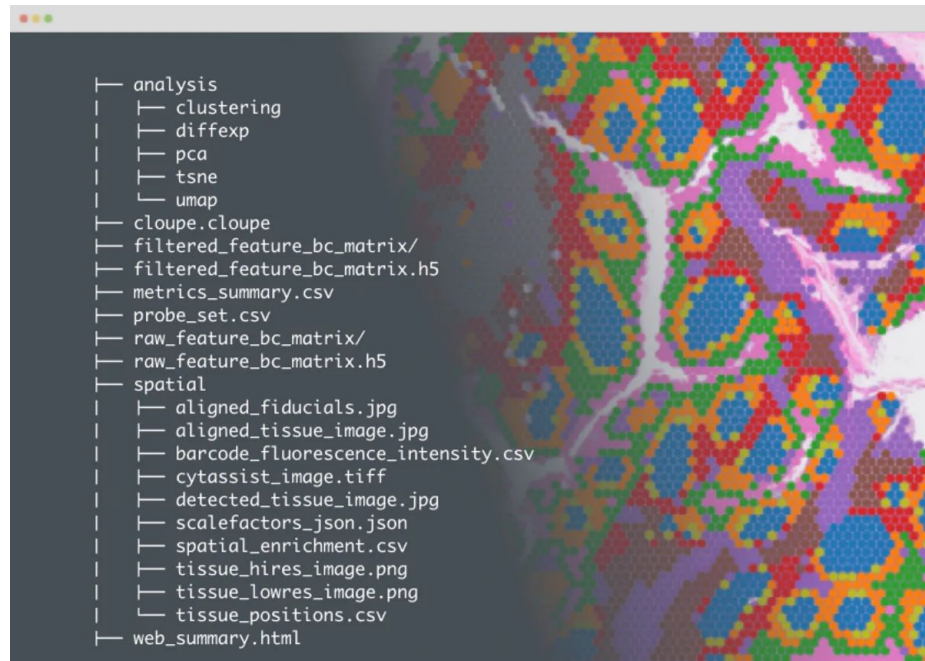
Space Ranger

Space Ranger helps to integrate the sequencing and imaging data

Space Ranger is a set of **analysis pipelines**

- **Count**
- Segment
- Annotate
- Aggr (aggregate)

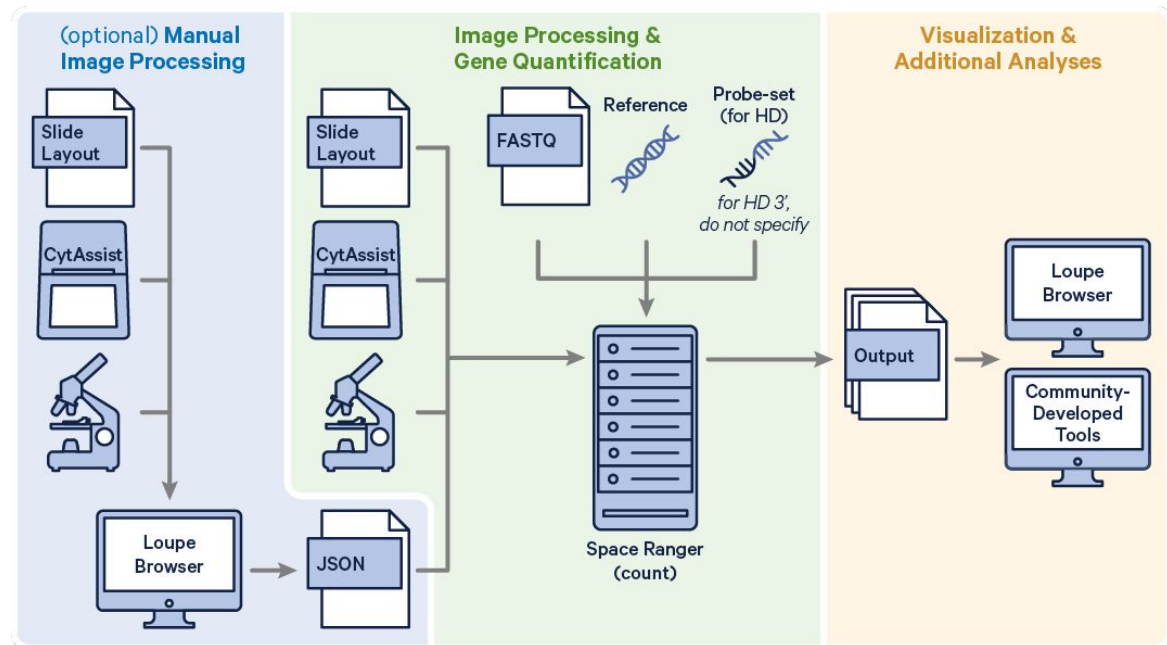
Allows local or cloud computing depending on resources



Space Ranger counts the no of transcripts and shows their location

Inputs:

- Reference genome
- Probe sets
- Slide layout files
- FASTQ files
- Microscope image
- Slide ID and Capture area



Outputs from Space Ranger

Here are few examples:

[Understanding Space Ranger Outputs | Official 10x Genomics Support](#)

[Space Ranger Web Summary | Official 10x Genomics Support](#)

[Space Ranger Secondary Analysis CSV | Official 10x Genomics Support](#)

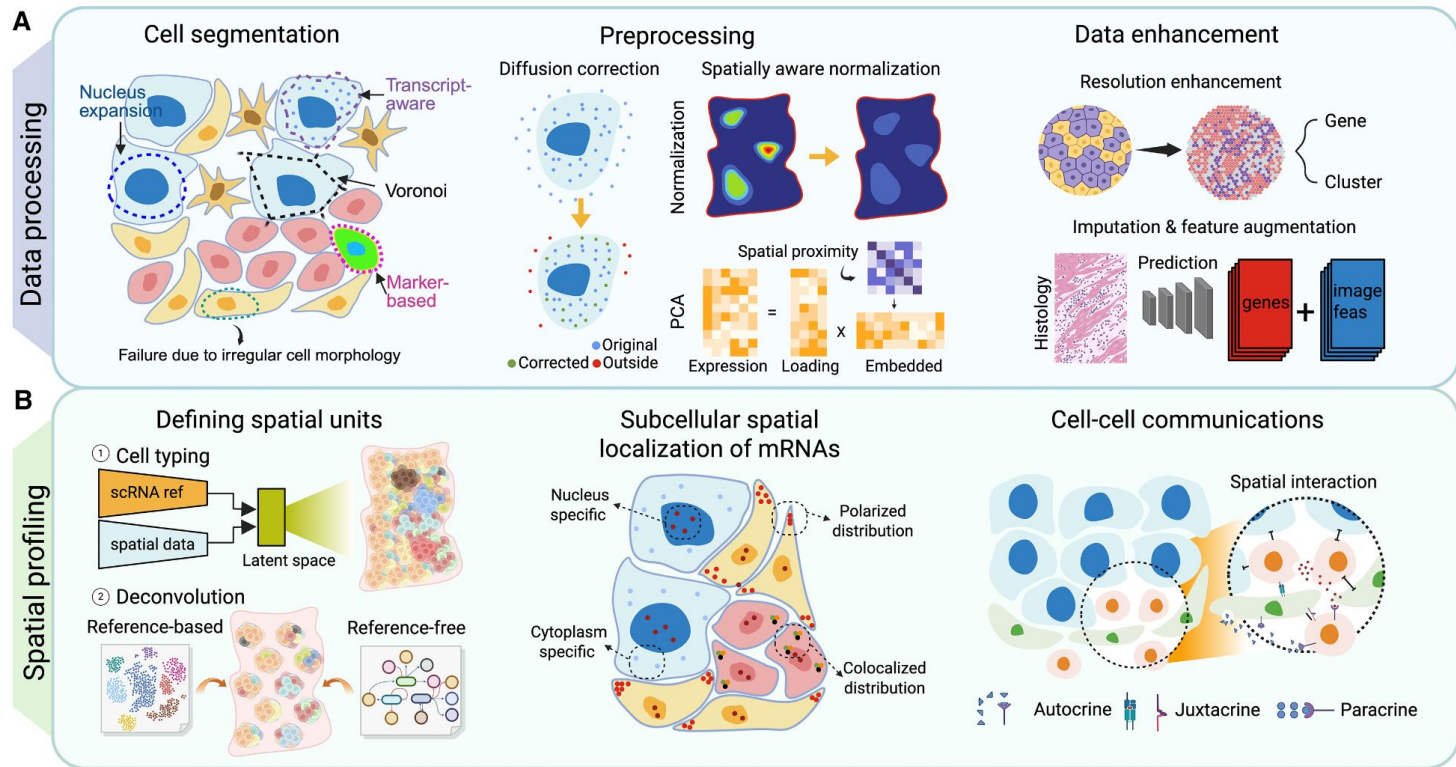
[Cell Type Annotation Outputs | Official 10x Genomics Support](#)

Public Datasets from 10x Genomics

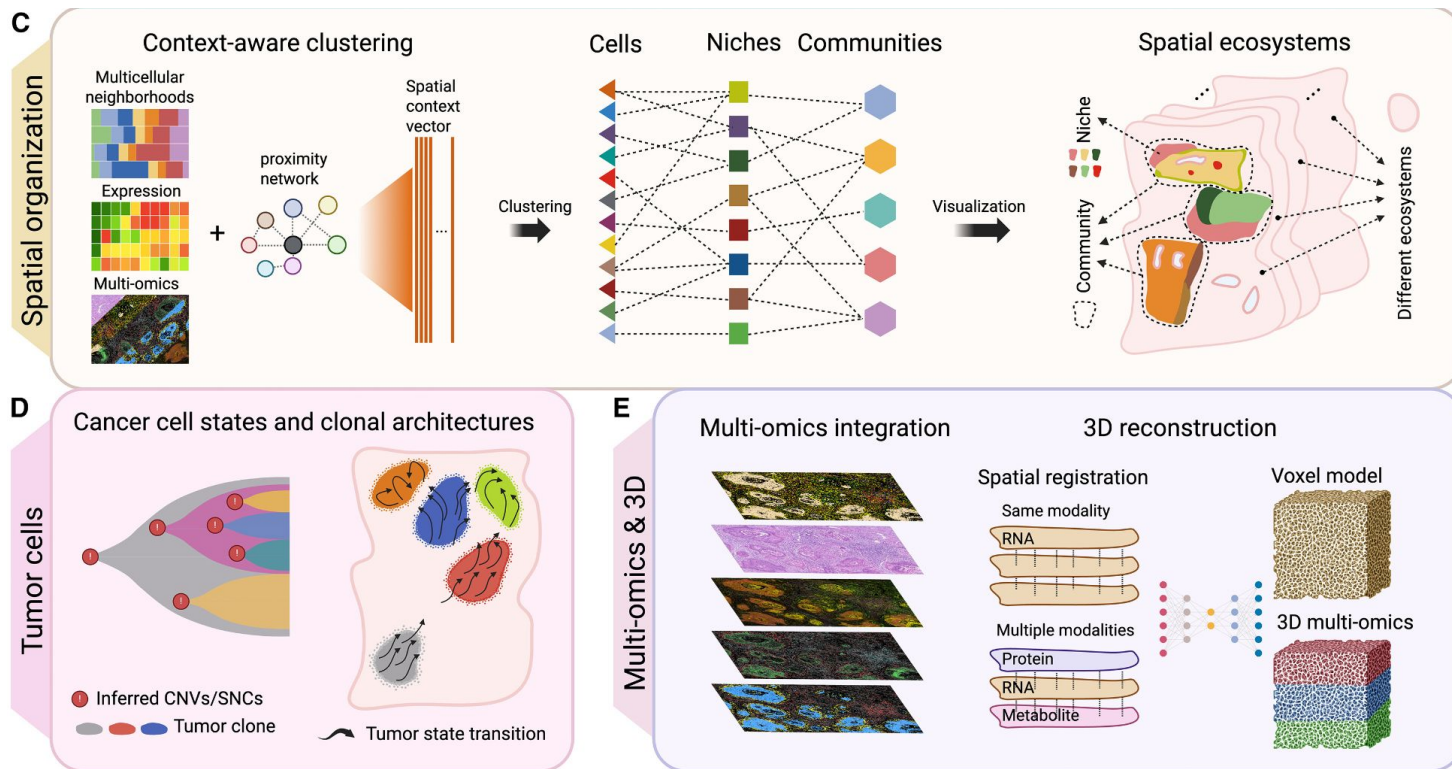
Datasets (Showing 48 datasets)	Product	Species	Sample type	Cells or nuclei	Preservation
Visium HD Spatial Gene Expression Library, Human Lung Cancer, Breast Cancer and Colon Cancer Tissue Array (FFPE), 11 mm Capture Area	HD Spatial Gene Expression v1 - Protocol 2.0	 Human	Lung, Breast, Colon	N/A	FFPE
Visium HD Spatial Gene Expression Library, Human Breast Cancer Tissue Microarray (FFPE), 11 mm Capture Area	HD Spatial Gene Expression v1 - Protocol 2.0	 Human	Breast	N/A	FFPE
Visium HD Spatial Gene Expression Library, Human Colon Cancer (FF), 11 mm Capture Area	HD Spatial Gene Expression v1 - Protocol 2.0	 Human	Colon	N/A	Fresh Frozen
Visium HD Spatial Gene Expression Library, Human Ovarian Cancer (FF) with Immunofluorescent staining, 11 mm Capture Area	HD Spatial Gene Expression v1 - Protocol 2.0	 Human	Ovary	N/A	Fresh Frozen
Visium HD Spatial Gene Expression Library, Mouse Embryo (FFPE), 11 mm Capture Area	HD Spatial Gene Expression v1 - Protocol 2.0	 Mouse	Embryo	N/A	FFPE
Visium HD Spatial Gene Expression Library, Human Colon Cancer (FFPE), 6.5 mm Capture Area	HD Spatial Gene Expression v1 - Protocol 2.0	 Human	Colon	N/A	FFPE

From Outputs to Insights

Some standard steps to process the spatial data



Some standard steps to process the spatial data



From closed to open source

Pros and Cons

Feature	Loupe Browser (10x Genomics)	Seurat-based R Workflow
User Interface	Click-based, interactive Desktop Graphical User Interface (GUI).	Code-based command-line interface (CLI) using R.
Learning Curve	Zero coding required. Highly intuitive for biologists.	Requires proficiency in R, data structures, and spatial packages.
Hardware Demands	Runs on standard desktops; handles pre-computed .cloupe files easily.	High RAM/CPU needed to process raw matrix files containing millions of bins.
Data Customization	Restricted to preset visualizations (UMAP/t-SNE overlay, basic violin plots).	Unlimited customization via ggplot2, custom color maps, and point sizes.
Advanced Statistics	Limited to built-in differential expression (e.g., Wilcoxon/edgeR variant).	Extensive options (SCTransform, Harmony batch correction, custom models).
Integration Power	Hard to integrate third-party multi-omics or multi-sample atlases natively.	Robust single-cell/spatial integration, anchor-based label transfer (Azimuth).
Reproducibility	Manual steps, region lassoing, and filtering are hard to script/track.	Code scripts provide absolute reproducibility and reproducible seeds.

R or Python

Criteria	scverse (Python Framework)	Seurat (R Framework)
Primary Data Structure	SpatialData object housing AnnData tables, Zarr arrays, and Geometries.	Seurat object containing multi-resolution assay slots + BPCells.
Large-Dataset Strategy	Lazy-Loading & Chunks : Uses Dask and Zarr to process large blocks only when queried.	Sketching & On-Disk Representation : Drops memory footprint by sampling or working off-disk.
Image Resolution Handling	Multi-scale pyramid images loaded dynamically depending on zoom level.	Loads downscaled images; struggles with full-resolution gigapixel raw H&E files natively.
Spatial Reference Integration	Advanced deep-learning reference mapping and graph neural networks (GNNs).	Excellent statistical label-transfer (Azimuth) and cell-type deconvolution.
Cell Segmentation	Handled cleanly via SpatialData Shapes components.	Natively supports spatial polygon subsets and zooming.

R based ecosystem



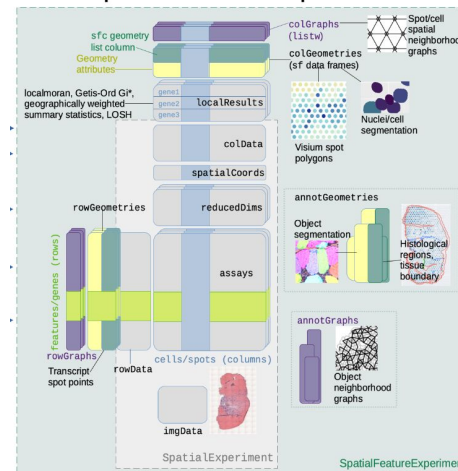
Rahul Satija
@satijalab



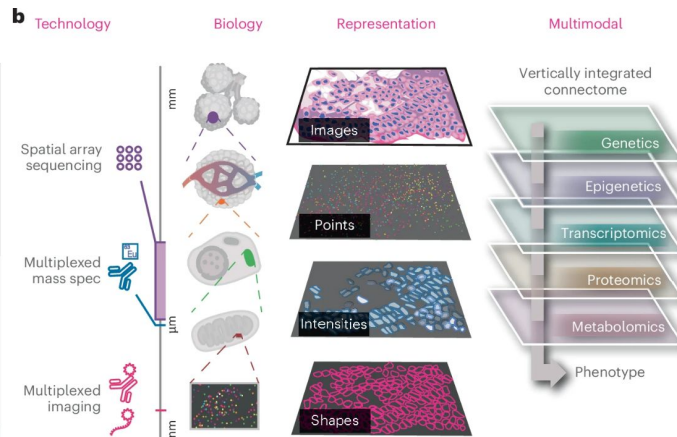
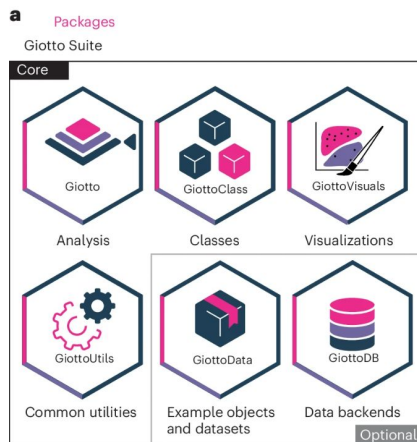
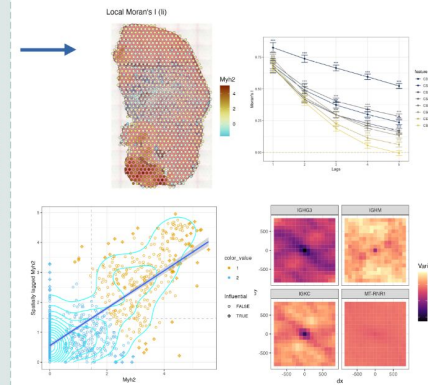
Over the past few months, we've been rewriting core workflows in Seurat with coding agents to improve speed and scalability. By moving steps from R to C++ and optimizing code, we've seen huge improvements - in some cases 10-fold and above - with identical outputs. (1/)

9:22 PM · Jun 22, 2026 · **29.7K** Views

SpatialFeatureExperiment



Voyager downstream analyses



Voyager: <https://doi.org/10.1101/2023.07.20.549945>

Giotto Suite: Chen, et al., *Nature Methods* (2025)

Seurat Architecture

Cell ID	nFeature	nCount	Percent_mito	Percent_ribo	...	droplet_status	Cell_type
Cell_1	18500	1000	17	2		singlet	99
Cell_2	16070	700	12	0		singlet	95
Cell_3	17780	980	5	1		singlet	92
...	18000	1600	25	5		doublet	89
Cell_50000	17070	2400	7	10		singlet	100

CELL
METADATA

DIMENSIONALITY REDUCTION DATA

Cell ID	PCA_1	PCA_2	...	UMAP_1	UMAP_2
Cell_1	5	5			
Cell_2	3.5	3.5			
Cell_3	0.78	0.78			
...	3	3			
Cell_500	1.2	1.2			

GRAPH DATA

Cell ID	Cell_1	Cell_2	Cell_3	...
Cell_1	1	1	1	1
Cell_2	1	1	1	1
Cell_3	1	1	1	1
...	0	0	1	1
Cell_50000	1	1	0	0

COUNTS TABLE

Gene ID	Cell_1	Cell_2	Cell_3	Cell_4	Cell_5	Cell_6	...	Cell_50000
TP53	80	50	63	36	60	0		99
EGFR	9	10	2.0	-2.3	-0.4	-4.3	-0.9	-9.4
VEGFA	10	6	3.0	-2.3	-0.4	-4.3	-0.9	-9.4
APOE	95	82	4.0	3.4	13.4	4.9	-4.9	-6.9
IL6	6	7	0.0	6.1	-5.3	4.1	-5.3	-4.3
TGFB1	82	70	2.0	4.1	8.1	2.6	-5.4	-2.6
AKT1	70	70	0.0	0.1	4.1	1.4	-6.1	0.1
MTHFR	70	70	0.0	2.3	3.7	1.7	-5.9	-4.3
...	70	70	0.0	0.7	8.3	2.1	-5.7	-4.1
BRCA1	70	70	1.0	0.6	4.4	3.4	-6.0	3.6
...	70	70	1.0	1.3	3.4	1.7	-6.4	10.9
BRCA1	70	70	1.0	1.3	3.4	1.7	-6.4	10.9

Raw counts

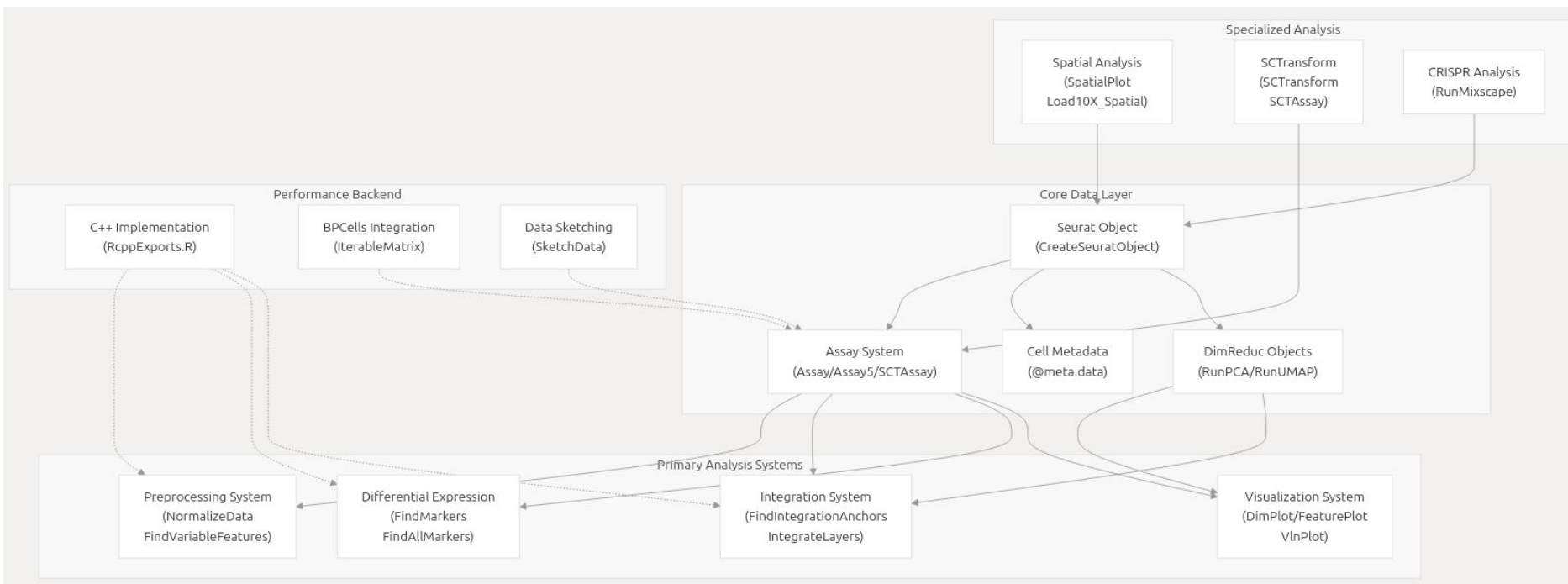
Normalised counts

Scaled counts

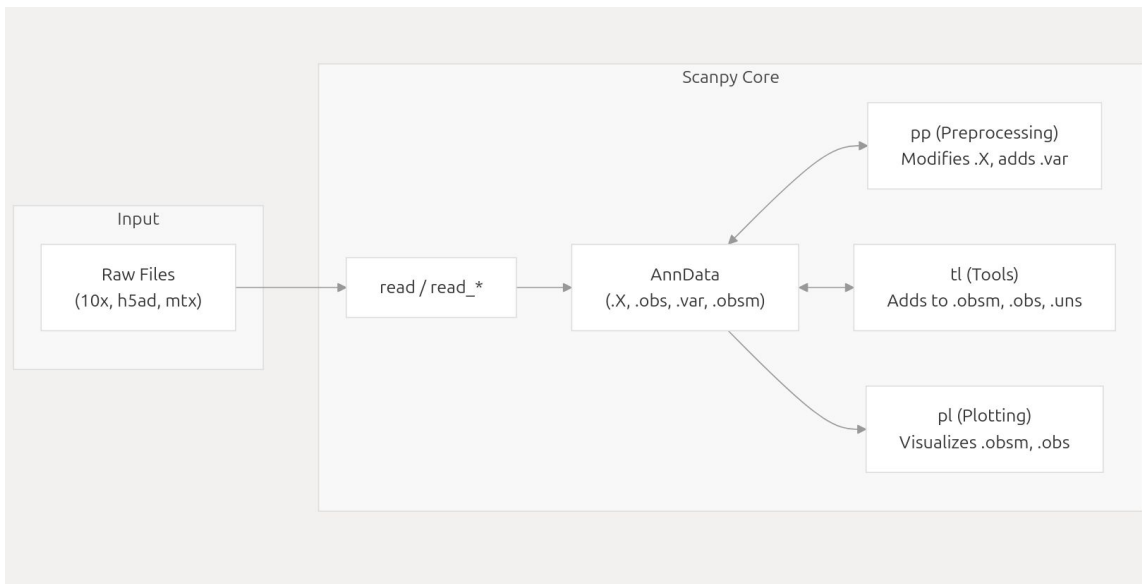
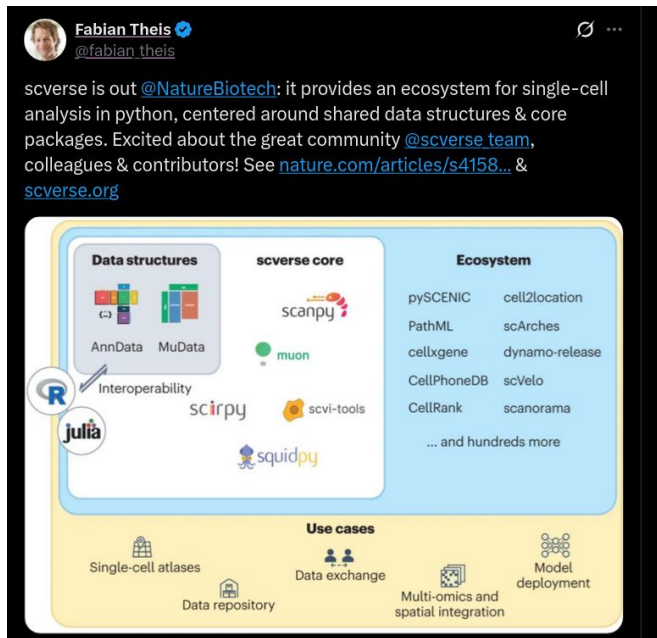
GENE METADATA

Gene ID	Avg_expression	...	variance
TP53	5		0.89
EGFR	3.5		1.2
VEGFA	0.78		2.2
...	3		0.76
BRCA1	1.2		2.4

Seurat Architecture



Python based ecosystem and architecture



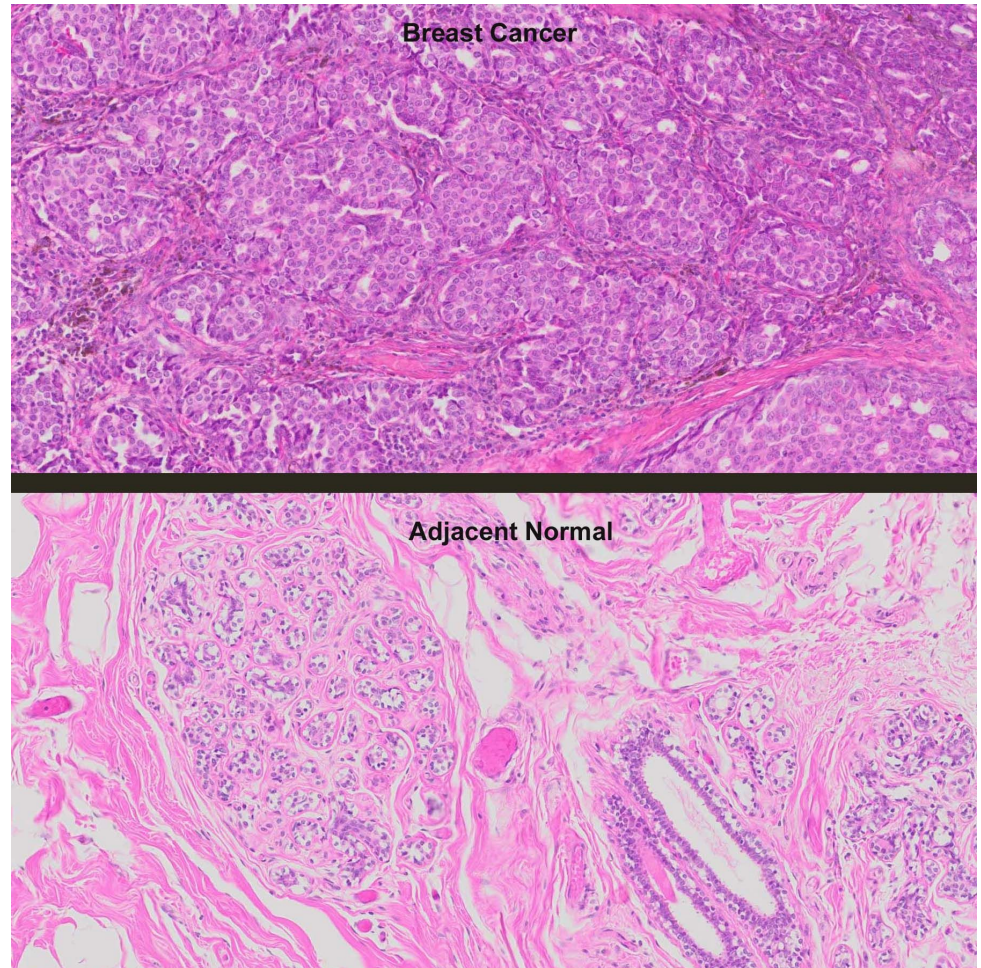
Cell Segmentation

H&E Staining

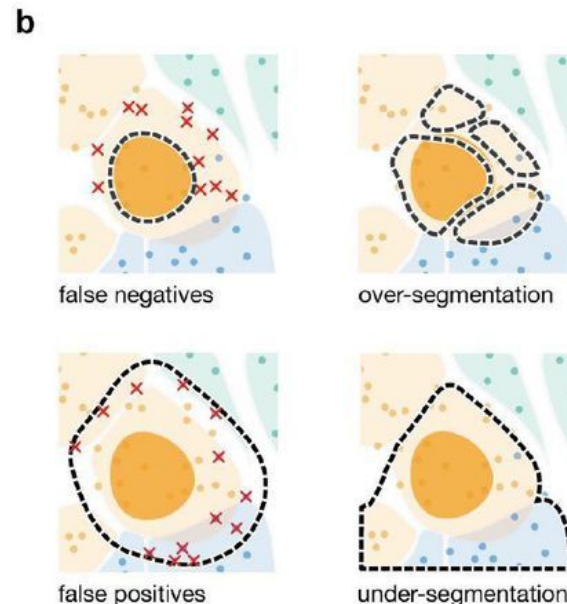
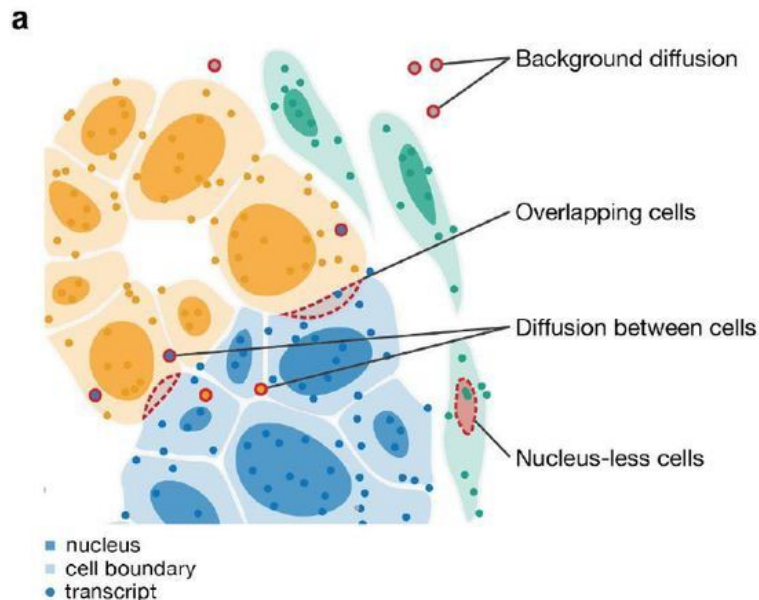
Hematoxylin (H) stains cell nuclei “purplish blue”

Eosin (E) stains the extracellular matrix and cytoplasm “pink”

Gold standard for medical diagnosis



Challenges in segmentation



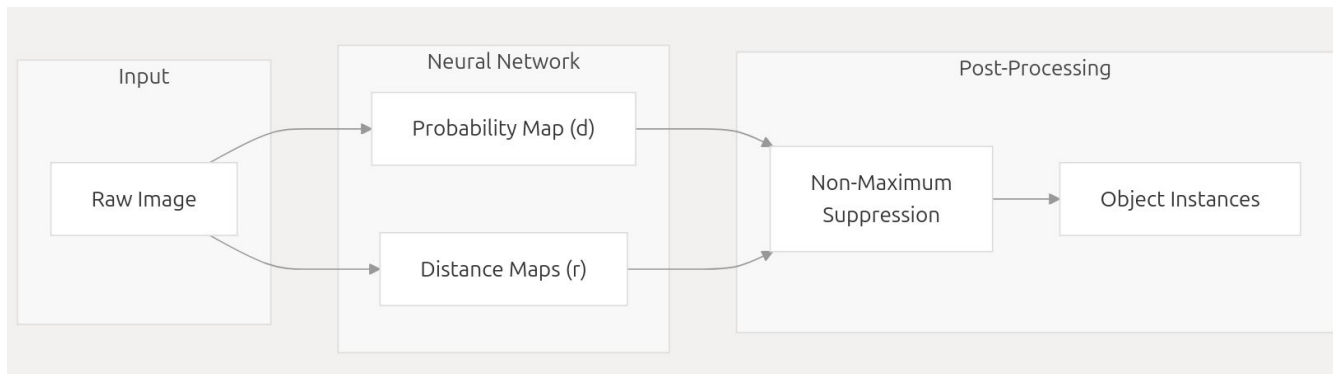
Essential for precise transcript mapping, accurate cell annotation, and cell-cell interactions

Segger: Fast and accurate cell segmentation of imaging-based spatial transcriptomics data
<https://doi.org/10.1101/2025.03.14.643160>

Image-Based Deep Learning Approaches

StarDist

- **Method:** Infer pseudo-boundary via nuclei expansion (DAPI/H&E).
- **Pros:** Compute fast. Handle distinct nuclei well.
- **Cons:** Miss true cytoplasm. Fail complex cell morphology.

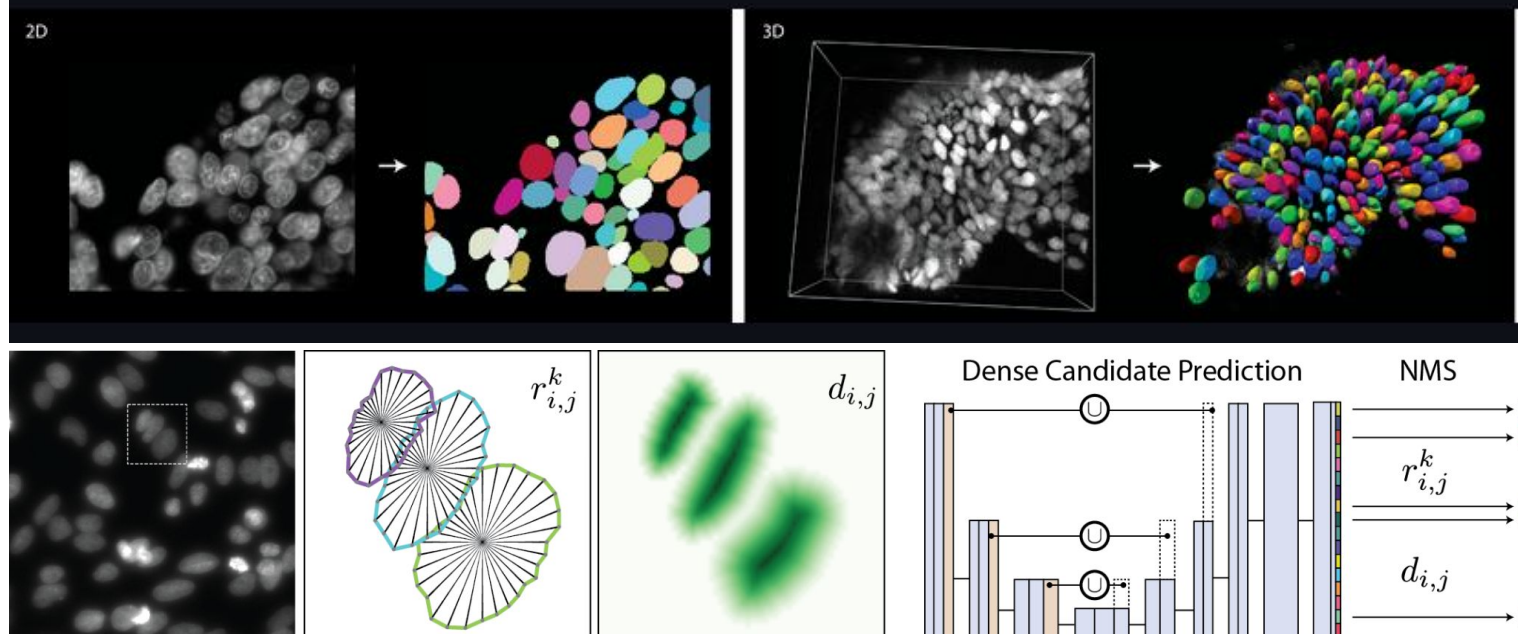


Cellpose / DeepCell

- **Method:** Predict true boundary using dual-channel signal (nucleus + membrane).
- **Pros:** High morphological accuracy. Capture non-convex shape.
- **Cons:** Require membrane stain. Struggle dense tissue overlap.

Image Segmentation

StarDist - Object Detection with Star-convex Shapes



Space Ranger v4.0+

Cell segmentation

Space Ranger v4.0 introduced cell segmentation for H&E images using the `spaceranger segment` pipeline, and for Visium HD data with H&E images using the `spaceranger count` pipeline. For cell segmentation in Visium HD analysis, Space Ranger utilizes a custom implementation of StarDist (Schmidt et al., 2018), a deep learning model well-suited for the instance segmentation of star-convex shapes, such as cell nuclei. StarDist uses a convolutional neural network (CNN) with a U-Net-like encoder-decoder architecture to extract relevant features from the input image patch. The result is a set of labeled instance masks, represented as star-convex polygons, delineating individual cells. StarDist was chosen because internal testing on a variety of human and mouse tissues demonstrated that StarDist outperformed comparable methods. In particular, StarDist excels at accurately separating closely clustered cells and segmenting cells in highly crowded environments due to its unique approach of predicting radial distances.

10x Genomics trained and benchmarked Space Ranger's StarDist v6 model in house on >17,000 image patches coming from >150 H&E tissue images. Preservation methods tested include FFPE, fresh frozen, and fixed frozen. Tissue types used for training, testing, and benchmarking include:

- Human: thymus, skin (melanoma), prostate, colon, colon cancer, breast, breast cancer, tonsil, thymus, brain, brain cancer, lung, lung cancer, and spleen
- Mouse: brain, bone, testes, small intestine, spleen, embryo, liver, lung, kidney, and thymus

Transcript-Aware & Probabilistic Approaches

Segger

- **Method:** Predict transcript-to-cell links using Graph Neural Network (GNN).
- **Pros:** Maximize transcript capture. High sensitivity. Resolve overlap. Low compute time.
- **Cons:** Depend on initial structural mask quality.

Baysor

- **Method:** Group transcripts via Markov Random Field probabilities.
- **Pros:** Stain-independent. Find cells via RNA density alone.
- **Cons:** Extreme compute cost. Scale poorly on large spatial datasets.

Current recommendation for Visium HD

- Run StarDist on tissue H&E image.

Bin2cell

2.3 Morphology and gene expression image segmentation

Nuclei are identified in the morphology image by using StarDist (Schmidt *et al.* 2018), an efficient segmentation algorithm that comes with pretrained H&E and fluorescence models. The nuclei labels are discretized into bins, and expanded into cells by label expansion of a fixed distance (by default 2 bins, i.e. 4 μm). Alternatively, approximating the cell and nuclei as concentric spheres, the linear relationship between their volume (Wu *et al.* 2022) can be used to propose a label-specific expansion distance d_L

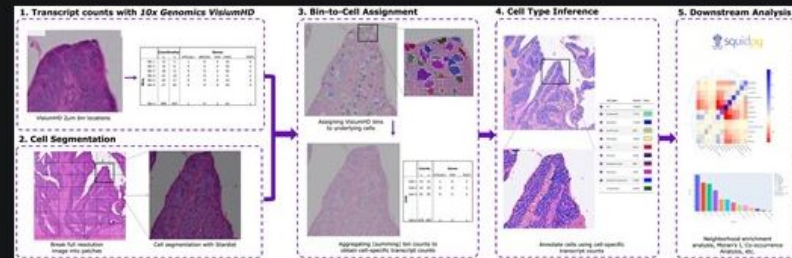
$$d_L = \left\lceil \left(\sqrt[3]{V} - 1 \right) \sqrt{\frac{n_L}{\pi}} \right\rceil,$$

where n_L is the number of bins labelled as nucleus L , and V is the cell-to-nucleus volume ratio with a default of 4 (Wu *et al.* 2022). Unassigned bins equidistant to bins from multiple nuclei are assigned based on transcriptional similarity, as represented in PCA space. An exploration of the two label expansion strategies with various parameter settings is performed in [Supplementary Section S5](#), with bin2cell performing robustly across the evaluation.

ENACT

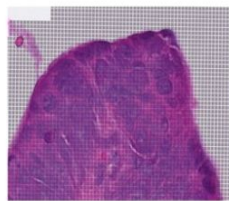
2.1 Cell segmentation

Figure 1 presents the key steps involved in the proposed processing and analysis pipeline. Cells in the full resolution tissue image are first segmented using Stardist, one of the most widely used UNet-based cell segmentation methods that has been shown to accurately segment images even when cells are tightly packed (Schmidt *et al.* 2018).



ENACT (Python based workflow)

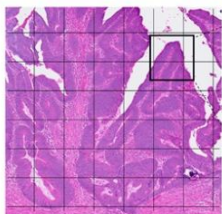
1. Transcript counts with 10x Genomics VisiumHD



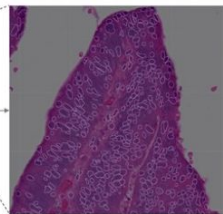
VisiumHD 2µm bin locations

	Coordinates		Genes				
	x	y	ATG16L1	BRISQ5	CD4	CD81	TGIF1
Bin 1	12	5	1	0	0	10	4
Bin 2	15	8	0	0	0	10	2
Bin 3	18	11	0	0	2	16	2
Bin 4	21	14	0	0	1	9	1
Bin 5	24	17	0	0	0	13	3
Bin 6	27	20	0	0	0	19	4
...	...	...	...	...	...	...	...
Bin n	900	457	1	0	2	14	1

2. Cell Segmentation

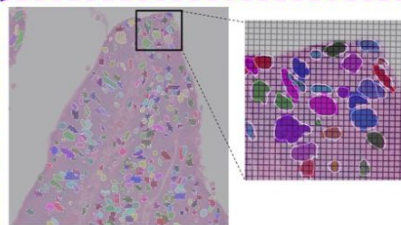


Break full resolution image into patches

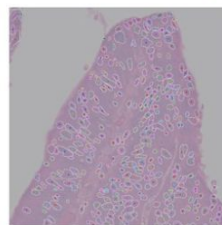


Cell segmentation with Stardist

3. Bin-to-Cell Assignment



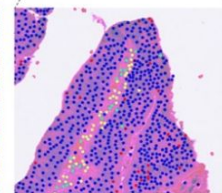
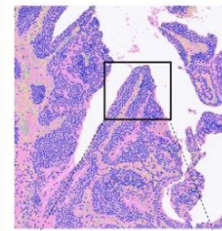
Assigning VisiumHD bins to underlying cells



Aggregating (summing) bin counts to obtain cell-specific transcript counts

	Coords.		Genes			
	x	y	ATG16L1	CD4	...	TGIF1
Cell 1	13	12	1	0	0	4
Cell 2	15	23	0	0	0	2
Cell 3	22	35	0	2	2	2
...	...	...	...	...	...	...
Cell n	670	697	1	2	1	1

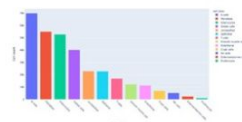
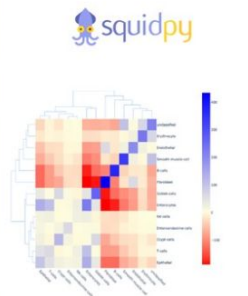
4. Cell Type Inference



cell_type	Counts	Color
All	378965	
Endothelial	17712	
Epithelial	123387	
Erythrocyte	908	
Fibroblast	76547	
HSC	4413	
Immune	9459	
Myeloid progenitor	894	
Neuronal	1605	
Smooth muscle cell	15397	
unclassified	94992	

Annotate cells using cell-specific transcript counts

5. Downstream Analysis



Neighborhood enrichment analysis, Moran's I, Co-occurrence Analysis, etc.

Cell Annotation

Reference Mapping Annotation

Seurat Label Transfer (Anchor-Based)

- **Method:** Find **Mutual Nearest Neighbors (MNN)** between Visium HD bins and scRNA-seq reference. Project cell-type labels via score matrix.
- **Pros:** **Native Seurat integration. Fast.** Standard for binned spatial data.
- **Cons:** **Assume linear relationship** between modalities. **Sensitive to spatial background noise.**



Azimuth

- **Method:** **Reference-mapping workflow.** Project Visium HD query onto curated, high-quality reference datasets (e.g., Allen Brain, HuBMAP Lung) using predefined transformation matrices.
- **Pros:** **Automated. Highly reproducible.** Standardize cell ontology across cohorts.
- **Cons:** **Limited to available public references.** Fail on novel tumor phenotypes.

Segmented Single-Cell & Bin-Level Deconvolution

BANKSY

- **Method:** Build feature matrix combining per-cell/bin expression with **local spatial neighborhood** context. Cluster combined matrix.
- **Pros:** Use tissue architecture. Identify microenvironment-driven cell states.
- **Cons:** Require **clean cell segmentation masks** or ultra-high resolution grid.

RCTD / Cell2location / Tangram

- **Method:** Probabilistic/**regression framework**. Fit scRNA-seq profiles to Visium HD spots/bins to estimate cell-type fractions.
- **Pros:** Deconvolve mixed signals in larger bins (8 μm /16 μm). **High statistical rigor.**
- **Cons:** **High compute cost** (Cell2location). Lose sub-bin spatial localization.

Spatial Context Phenotyping

Spatial-ID / STANN

- **Method:** Deep learning / transfer learning frameworks. Map scRNA-seq features to spatial single-cell data using spatial embeddings.
- **Pros:** High cell-typing accuracy. Handle low gene coverage well.
- **Cons:** Black-box neural net models. Need massive training data.

ScType

- **Method:** Reference-free annotation. Filter cell-specific marker genes against curated signature databases to compute cell-type enrichment scores.
- **Pros:** Zero scRNA-seq reference dataset requirement.
- **Cons:** Fail if marker genes lack specificity or drop out in spatial transcripts.

Spatially Variable Genes

Spatially Variable Genes

Goal

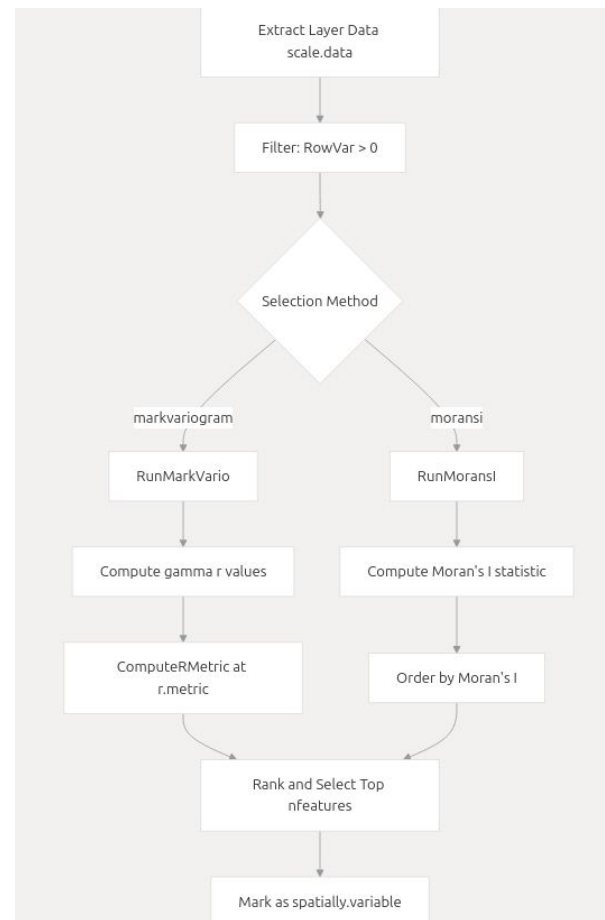
- SVG show non-random tissue location. Find **biological structure**.

Moran's I

- To measure **spatial autocorrelation**. Evaluate spot vs neighbor expression.
- Score +1 (cluster). Score 0 (random). Score -1 (disperse).
- Seurat call: `method = "moransi"`.

Markvariogram

- To measure **spatial variance over distance**. Find dependency range.
- Spot expression difference grow as physical distance grow.
- Seurat call: `method = "markvariogram"`.

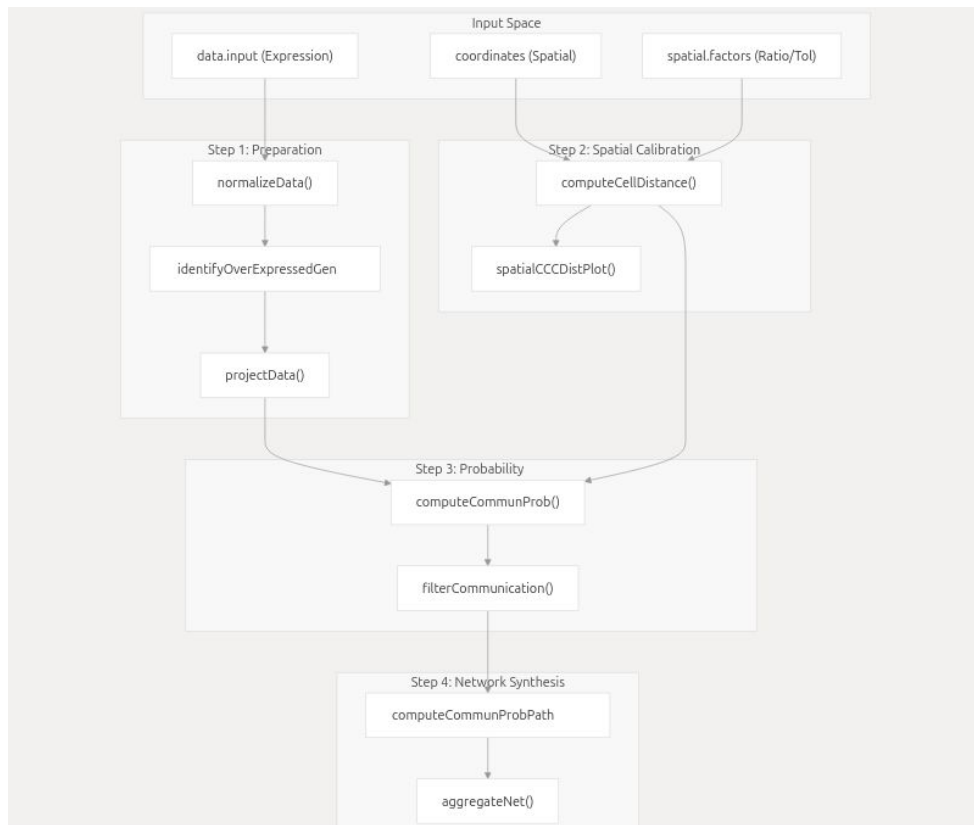


Ligand Receptor Interactions

Ligand Receptor Interactions

SpatialCellChat

- Measures physical cell-centroid distance.
- Filter drop pairs beyond maximum diffusion limit.
- Model combine ligand-receptor expression with spatial proximity.
- Hill function compute communication probability.
- Output rank active signaling pathways.



SpatialCellChat Workflow

`createSpatialCellChat()`:
Initialize object with
expression data, metadata,
coordinates, and spatial
factors

Set CellChatDB: Load and
subset ligand-receptor
interaction database

`subsetData()`: Subset
expression data to
signaling genes only

`preProcessing()`: Apply
ALRA denoising and PPI-
based smoothing

`identifyOverExpressedGenes()`:
Identify spatially variable
ligands/receptors using
methods like meringue

`identifyOverExpressedInteractions()`:
Identify significant L-R pairs where
either ligand or receptor is over-
expressed

`computeCommunProb()`:
Model communication
probability using law of
mass action with spatial
distance constraints

`filterProbability()`: Filter
out statistically non-
significant communications

`filterCommunication()`:
Remove communications
with few interactions

`computeAvgCommunProb_Visium()`:
Aggregate probabilities at cell
group level for low-resolution data

`computeCommunProbPathway()`:
Summarize probabilities at
signaling pathway level

`aggregateNet()`: Calculate
aggregated network by
counting links and
summarizing probabilities

`netAnalysis_computeCentrality()`:
Compute network centrality
scores to identify major senders/
receivers

Ligand Receptor Interactions

COMMOT

- **Method:** Optimal Transport model cost-efficient ligand flow to receptors.
- **Pros:** Map continuous spatial gradients. High math rigor.
- **Cons:** Compute heavy. Assume steady-state flow.

NCEM (Node-Centric Expression Models)

- **Method:** Graph Neural Network predict target transcript changes from neighbor states.
- **Pros:** LR database independent. Link neighborhood to expression directly.
- **Cons:** Pure statistical correlation. Lack strict molecular mechanism.

NicheNet / SpaTalk

- **Method:** Model trace secreted ligands to downstream target gene activation.
- **Pros:** Prove functional outcome (pathway activity). Confirm true signal reception.
- **Cons:** Demand high gene coverage. Drop out on sparse data.

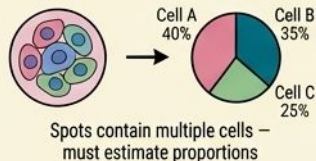
Spatial Transcriptomics

Seeing is believing, in an ideal world

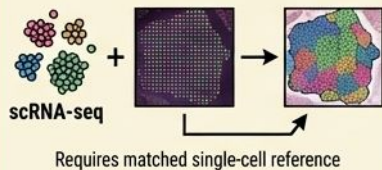
COMPUTATIONAL CHALLENGES IN SPATIAL TRANSCRIPTOMICS

Low-Resolution Sequencing

Challenge 1: Deconvolution



Challenge 2: Reference Integration



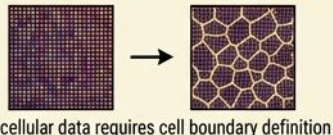
Tools: cell2location, RCTD, SPOTlight, Tangram

Difficulty

Medium

High-Resolution Sequencing

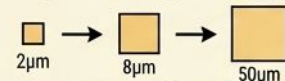
Challenge 1: Cell Segmentation



Challenge 2: Scalability



Challenge 3: Binning Strategy

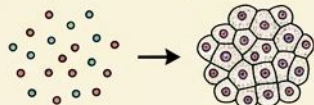


Must choose resolution vs. data depth trade-off

Tools: Cellpose, Baysor, Squidpy, SAW
High

Imaging-based

Challenge 1: Cell Segmentation



Challenge 2: Assignment Errors



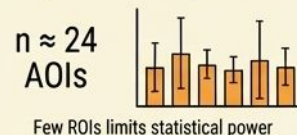
Challenge 3: Panel Limitations



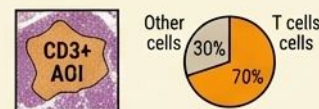
Tools: Cellpose, Baysor, StarDist
High

ROI-based

Challenge 1: Low Sample Size



Challenge 2: Cell Mixtures



Tools: SpatialDecon, GeoMx software

Difficulty

Low-Medium (computation)
High (statistics)

ALL PLATFORMS



Batch Effects

Technical variation between samples
Harmony, scVI



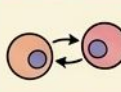
Normalization

Sequencing depth correction
SCTransform



Spatial Statistics

Spatially-aware statistical tests
Squidpy, SpatialDE



Cell-Cell Interactions

Infer cell communication
CellChat, CellPhoneDB

Experimental challenges in spatial transcriptomics

Cell Clumping

- **Problem:** Tissue dense. Cells overlap z-axis.
- **Impact:** Mask algorithms fail irregular borders. Single-cell transcripts mix.
- **Next Step:** Train 3D membrane deep learning models.

RNA Diffusion

- **Problem:** Permeabilization leak mRNA. Transcripts float away.
- **Impact:** Neighbor cells receive false positive transcripts. True biological signal mix with technical bleed.
- **Next Step:** Improve probe chemistry. Run diffusion-aware imputation algorithms.

Scope for new methods



sina @sinaboreshaghi · Jun 11

2. In a new preprint (w @IngileifBryndis & @agalvezmerchan, and @lpachter), we show that PFlogPF is the normalization method that best satisfies the following three requirements for single-cell RNA-seq data analysis:

- depth normalized
- variance stabilized
- monotonic

The centered log-ratio transform (CLR) was first proposed in (Aitchison, 1982), and the shifted centered log-ratio transform can be understood as an extra proportional fitting step after the standard log1p transform. Specifically, for a cell with counts $\mathbf{x} = (x_1, \dots, x_D)$, let

$s = \sum_{j=1}^D x_j$ and $u_j = x_j/s$. The PfllogPF family applies a shifted log and then centers in log space:

$$z_i = \log(u_i + c) - \frac{1}{D} \sum_{j=1}^D \log(u_j + c),$$

where $c > 0$ is the pseudocount. In the benchmark, we use $c = 1$, and refer to this method as PFlog1pPF (CLR).

Shifted centered-log ratio (CLR) transform

Method	Monot.	Add.	Equiv.	Scale inv.
PFlogPF (CLR)	✓	✓	✓	✓
$\log(y/s + 1)$	✓	×	✓	✓
$\text{acosh}(2\alpha y/s + 1)$	✓	×	✓	✓
$\log(y/s + 1/(4\alpha))$	✓	×	✓	✓
$\log(\text{CPM} + 1)$	✓	×	✓	✓
$\log(y/s + 1)/u$	✓	×	✓	✓
$\log(y/s + 1) \rightarrow \text{HVG}$	—	—	—	—
$\log(y/s + 1) \rightarrow Z$	×	×	✓	✓
$\log(y/s + 1) \rightarrow \text{HVG} \rightarrow Z$	—	—	—	—
Pearson	×	×	✓	×
Pearson (no clip)	×	×	✓	×
Analytic Pearson	×	×	✓	×
sctransform	—	—	—	—
Random quantile	—	—	—	—
Pearson $\rightarrow$ HVG	—	—	—	—
Pearson $\rightarrow Z$	×	×	✓	×
Pearson $\rightarrow \text{HVG} \rightarrow Z$	—	—	—	—
Sanity MAP	—	—	—	—
Sanity distance	—	—	—	—
Dino	—	—	—	—
Normaliser	—	—	—	—
GLM PCA	—	—	—	—
NewWave	—	—	—	—

Table 1: Axiom check for PFlogPF (CLR) and the 22 transformations benchmarked in (Ahlmann-Eltze and Huber, 2023), excluding the natural-log calibration axiom. The columns correspond to rank monotonicity, perturbation additivity, relabeling equivariance, and scale invariance. The table assesses the transformations as benchmarked. A dash indicates that the method is not an ordinary map $\mathbb{R}_{>0}^D \rightarrow \mathbb{R}^D$ to which the axioms directly apply, for example because it includes feature selection, randomization, distances, or low-dimensional model fitting.

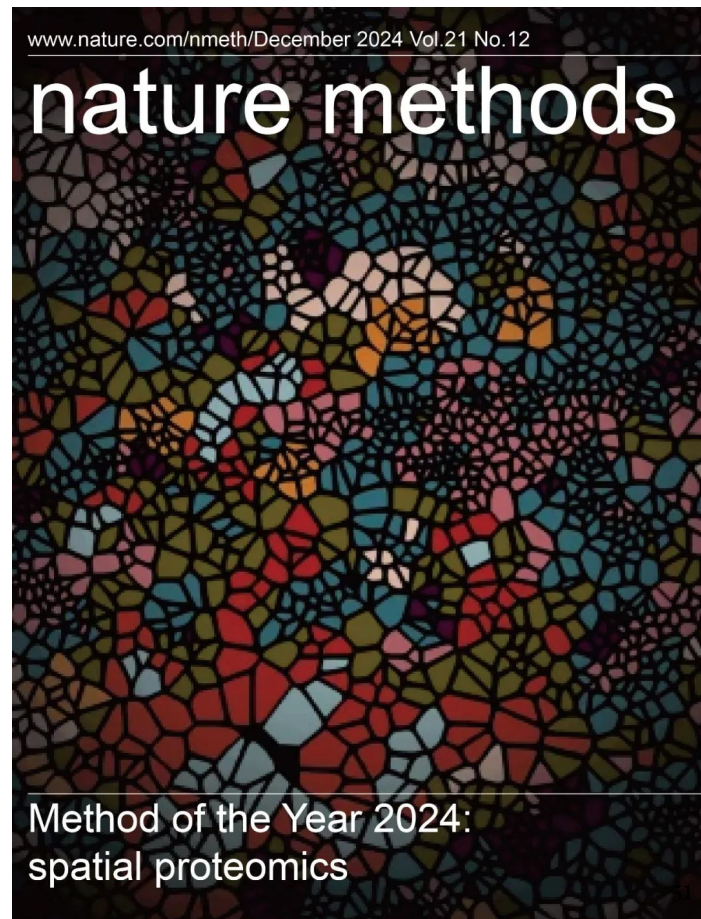
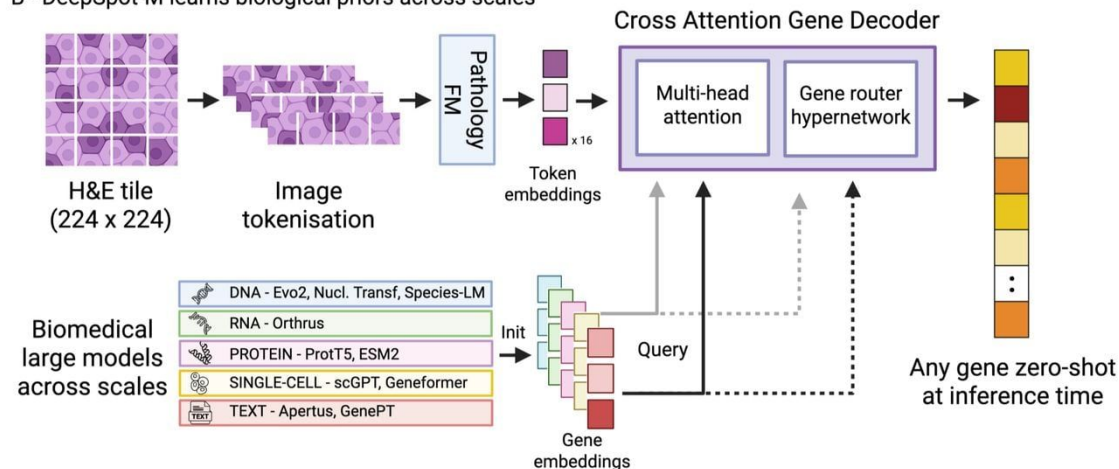
Future opportunities

DeepSpot-M: a multimodal foundation model for transcriptome-wide virtual spatial transcriptomics from histology

 Kalin Nonchev, Sebastian Dawo, Karina Silina,  Viktor H Koelzer, Gunnar Rätsch

doi: <https://doi.org/10.64898/2026.06.19.26356060>

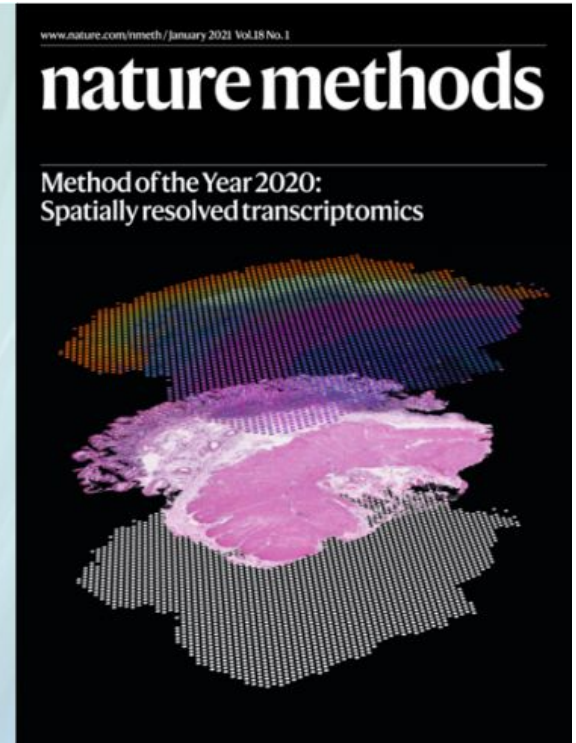
B - DeepSpot-M learns biological priors across scales



Read journals to know what's going on...



Method of the Year 2019: Single-cell multimodal omics

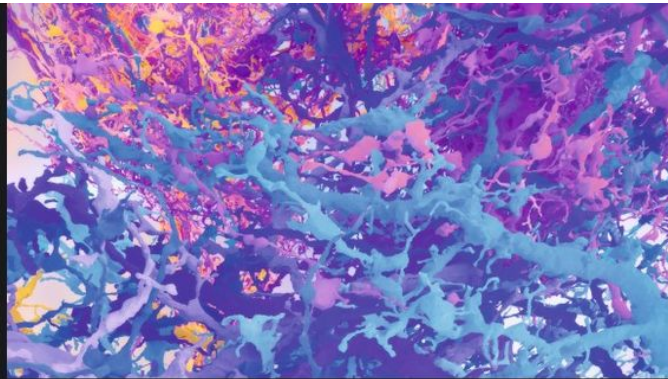


Guess the method of the year for 2025

Focus | 08 December 2025

Method of the Year 2025: EM-based connectomics

Our choice for the Method of the Year 2025 is EM-based connectomics, for the detailed insights into brain structure and function that the generated wiring diagrams enable.



Guess the method of the year for 2026

Additional Resources

1. <https://github.com/ashoks773/SpatialTranscriptomicsWorkflow> (Python + R)
2. https://satijalab.org/seurat/articles/visiumhd_analysis_vignette (R based)
3. https://hbctraining.github.io/spatial_nanocourse/lessons/visium_hd.html (R based)
4. https://github.com/sheffield-bioinformatics-core/spatial_transcriptomics_tutorial (R based)
5. <https://omicstutorials.com/spatial-transcriptomics-capturing-the-spatial-organization-of-gene-expression/> (Theory)
6. <https://www.epfl.ch/research/facilities/gene-expression-core-facility/wp-content/uploads/2026/06/Guidelines-10XG-Visium-HD-probes-by-GECE-vA.09.pdf> (Experimental guideline)
7. https://cdn.10xgenomics.com/image/upload/v1756505813/support-documents/CG000805_VisiumHD3_UserGuide_RevB.pdf (10x user guide)
8. Grases D, Porta-Pardo E, "A practical guide to spatial transcriptomics: lessons from over 1000 samples", Trends in Biotechnology, 2025; 44, 1230-1242 (Best practices)
9. Liu, Yunhe, et al. "Spatial Omics at the Forefront: Emerging Technologies, Analytical Innovations, and Clinical Applications." *Cancer Cell*, vol. 44, no. 1, 2026, pp. 24-49, <https://doi.org/10.1016/j.ccell.2025.12.009>. (Comprehensive review of spatial technologies)

Feel free to contact me
for more ideas/questions

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